

Title: Design and Prototype of an Innovative Soap/Detergent Product

Objective:

- Foster creativity and problem-solving in developing a functional and sustainable detergent or soap.
- Apply scientific principles in surfactant chemistry, formulation, and product testing.
- Encourage collaboration, prototyping, and innovation in the development of hygiene products.
- Promote awareness of sustainability, emerging trends, and consumer safety in the detergent industry.

Assignment Scope

- **Design and produce a prototype soap or detergent** (solid, liquid, or powder form).
- **Select appropriate ingredients** and justify their functionality (e.g., surfactants, builders, fillers, fragrances, coloring, thickeners, preservatives).

Address key considerations such as:

- Cleaning performance
- Skin-friendliness or surface compatibility
- Biodegradability and environmental safety
- Cost-effectiveness and scalability
- Prepare **product labeling and packaging** that complies with hygiene and safety standards.
- Conduct **basic testing** of the product (e.g., pH, foaming, emulsification, cleaning power).

Deliverables:

Prototype Sample (small quantity for demonstration)

Product Design Report (8-10 pages) including:

- Introduction and product concept
- Ingredient list and formulation rationale
- Preparation method and process
- Test results and performance evaluation
- Biodegradables and Toxicity assessment
- Sustainability and safety considerations
- Reflection on team collaboration and learning

Marketing Poster or Label showcasing

- Product name and brand
- Instructions for use
- Safety precautions
- Sustainability claims (if any)